Mastering the Craft

AI-Powered Intelligent Tutoring for Speedrunning

Bridging the gap between passive learning and elite execution through real-time AI coaching.

> *Speedrunning isn't just a game; it's an optimization problem.*
> *We built the first local AI coach to help players solve it.*

Presenter
Charlie Miner

The Problem: The "Learning Wall"

The Complexity

Minecraft 1.16 is a "multimodal" challenge:

- 3D navigation & pathing
- Real-time resource management
- High-pressure mechanical execution

The Current Gap

Passive Learning
Watching pros provides knowledge but zero feedback loop.

Trial & Error
Inefficient and discouraging; players don't know *why* they failed.

> *The Mission: Turn frustrating "dead runs" into actionable learning opportunities.*

Related Work

Feature	YouTube/Twitch	Metafy Guide	Mastering The Craft
Analysis Depth	Surface/Visual	Expert Lecture	Granular Telemetry
Feedback Loop	N/A	Static/Manual	Real-time & Interactive
Learning Mode	Observational	Procedural/Strategic	Dialogue and Practice Based

> *Closing the execution gap where passive observation fails.*

Learning Components

Knowledge

Focuses on technical resource minimums and **optimizing** strategic routing paths.

Cognitive

Developing **problem-solving** under extreme pressure and identifying critical moments to pivot strategies.

Affective

Reducing **"RNG frustration"** by shifting focus to celebrate incremental mechanical growth.

AI Architecture & Implementation

The Engine: Ollama

Llama 3.2 running locally for maximum privacy and zero latency during practice sessions

Expert System Prompting

Encoded "Speedrun Lore" including optimal split times and resource management rules.

Contextual Analysis

Parses JSON telemetry to infer Player Intent versus Actual Execution in real-time.

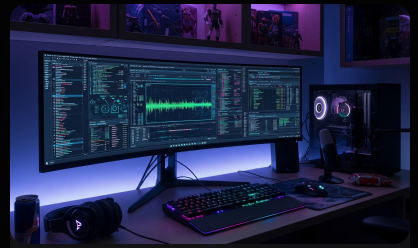
Adaptive Feedback

Adjusts explanation depth dynamically based on player inquiries and skill level.

The Learner Experience

1. Preparation

Set up Ollama and point a file path to get ITS set up.



2. Active Practice

Player plays a seed while the system tracks inventory, movement, and crafting telemetry.

3. Coaching Report

AI compares run against Expert Baseline to identify the "Top 3 Time Losses."

4. Dialogue

Interactive follow-up questions for deep mechanical breakdowns and strategy.

> *A seamless feedback loop designed to bridge the gap between intent and execution.*

Key Takeaways

AI as a Force Multiplier

Moving from basic chatbots to domain-specific experts that understand the nuances of elite performance.

Actionable Data

Turning raw telemetry into a constructive conversation, bridging the gap between numbers and execution.

Scalability

A blueprint for coaching in any high-skill, data-rich environment, from gaming to professional technical fields.

```
📄 Analysis data saved to: /Users/charlie_miner/.speedrun_tutor_last_analysis.json
   View with: cat /Users/charlie_miner/.speedrun_tutor_last_analysis.json
\nAnalyzing your run...\n
RUN SUMMARY:
The runner started by attempting to gather resources in the overworld, but quickly fell be
hind schedule due to inefficient farming and portal placement.

TOP TIME LOSSES:
1) Over-farming at 8:45 resulted in a 30-second delay, costing valuable time.
2) Portal entry was delayed by 4 seconds after finding iron, leading to additional time lo
st.
3) In the Nether, getting 7 Blaze rods took 52 seconds, exceeding the ideal time.

TOP FIXES FOR NEXT RUN:
1) Optimize farming route to reduce over-farming time.
2) Improve portal placement and entry strategy to shave off precious seconds.
3) Increase speed in finding and collecting Blaze rods.

PRACTICE DRILL:
Find a suitable location for a quick 20-second Blaze rod collection drill.
\nYou can now chat with the tutor. Type 'exit' to quit.\n
You:
```